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Assessing AgERA5 and MERRA-2 Global Climate Datasets for Small-Scale Agricultural Applications †

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Abstract: AgERA5 (ECMWF) is a relatively new climate dataset specifically designed for agricultural applications. MERRA-2 (NASA) is also used in agricultural applications; however, it was not specifically designed for this purpose. Despite the proven value of these datasets in assessing global climate patterns, their effectiveness in small-scale agricultural contexts remains unclear. This research aims to fill this gap by assessing the suitability and performance of AgERA5 and MERRA-2 in precision irrigation management, which is crucial for regions with limited ground data availability. The wine-making region of Nemea, Greece, with its complex and challenging terrain is used as a characteristic case study. The datasets are assessed for key weather variables and for irrigation planning, using detailed local meteorological station data as a reference. The results reveal that both products have serious limitations in small scale irrigation scheduling applications in contrast to what was reported in previous studies for other regions. The uneven performance of global datasets in different regions due to lack of sufficient observation data for reanalysis data calibration was also indicated. Comparing the two datasets, AgERA5 outperforms MERRA-2, especially in precipitation and reference evapotranspiration. MERRA-2 shows comparable potential in irrigation planning, as it occasionally matches or exceeds AgERA5's performance. The study findings underscore the importance of evaluating metanalysis datasets in the application area before their use for precision agriculture, particularly in regions with complex topography.

Keywords: global climate datasets; AgERA5; MERRA-2; viticulture; precision irrigation management; agricultural water management; agrometeorological data

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1. Introduction

In regions with limited meteorological data, effective irrigation management is challenging, necessitating alternative data sources such as global climate datasets. Moreover, the increasing prevalence of water scarcity, coupled with agriculture's position as the largest consumer of water, underscores the importance of optimum water use in agriculture, namely, determining the appropriate timing and quantity of irrigation [1].

Accurate representation of climate variables in numerical atmospheric models is essential for understanding the physical mechanisms driving the hydrological cycle and predicting future changes [2]. Precipitation, temperature, and other meteorological parameters significantly impact socioeconomic conditions, influencing agriculture, water resources, and disaster management [3]. In agriculture, reliable meteorological data are crucial for effective agricultural management practices, such as irrigation scheduling, which directly affects crop production [4]. This is particularly true for small-scale agricultural areas that face unique challenges, including data scarcity and limited resources for infrastructure development [2]. Global climate datasets, like reanalysis products, can be proven imperative in mitigating these limitations [5–8].

Reanalysis datasets are generated by assimilating various observational data sources, including satellite data, weather stations, and other remote or proximal sensing data, into numerical weather prediction models [9]. These datasets, such as the ERA5 from the European Centre for Medium-Range Weather Forecasts (ECMWF) and MERRA-2 from the National Aeronautics and Space Administration (NASA), offer global spatial and extensive temporal coverage, making them valuable for agricultural applications [10]. The high-resolution data provided by these reanalysis products create new possibilities for ungauged regions, such as better agricultural management decision-making, particularly in irrigation scheduling.

Despite their benefits, there are still significant uncertainties in existing atmospheric models. These uncertainties arise from the complex nature of weather processes and the limitations of model physics, which can be sensitive to small changes in input data [10]. Different reanalysis products may yield varying results due to differences in data assimilation techniques and input data sources [7,11]. Therefore, it is essential to evaluate and compare these datasets to determine their accuracy and applicability in specific contexts [12,13].

Irrigation scheduling involves determining the optimal timing and amount of water that should be applied to crops to maximize water use efficiency and crop yield. Proper irrigation management is critical in water-scarce regions, as it allows for significant water savings and at the same time it enhances crop health and productivity [14,15]. Especially for vineyards in Mediterranean countries, such as those in Nemea, Greece, efficient irrigation scheduling is crucial due to the water scarcity and the coincidence of the crop growth period with the dry period resulting in high crop water requirements. However, precision irrigation management is challenging as it requires accurate and up-to-date data for a wide range of parameters. The main factors that should be considered in proper irrigation scheduling are the soil hydraulic characteristics, the crop characteristics, the characteristics of the irrigation system, and most important, the meteorological conditions [16]. Among them, accurate weather data are highly important to estimate crop water requirements and to create detailed irrigation plans [17,18]. AgERA5 is a relatively new climate dataset specifically designed with agricultural applications in mind. It is derived from the established ERA5 dataset produced by ECMWF but offers a refined and tailored version focusing on variables and parameters most relevant to agricultural needs combined with a finer spatial resolution and daily aggregated time step [19]. MERRA-2 is a comparable dataset that provides valuable data for various hydrological applications. While MERRA-2 is utilized in agricultural applications, it was not specifically developed to support agricultural needs [20–22].

A limited number of studies have explored the potential of reanalysis datasets in large-scale agricultural applications, and particularly in agricultural water management. For instance, previous studies evaluating ERA5 for global or large-scale agricultural water management applications showed good agreement with real data [23] or in some cases acceptable errors [24], supporting the model's potential for agricultural applications in

regions with limited meteorological data. Additionally, Vanella et al. [25] in a study comparing ERA5 and ERA5-Land reanalysis datasets with ground-based agrometeorological data across diverse climates in Italy also observed a good agreement for variables such as air temperature and relative humidity, with ERA5-Land showing slightly higher accuracy for estimating reference evapotranspiration. Similarly, Lorite et al. [26] in their study investigating the use of weather forecast models for irrigation scheduling found minimal differences between the use of forecasted and observed data in irrigation water depths and crop yields. However, the performance of reanalysis datasets can vary temporally. Li et al. [10] for instance, evaluating four datasets (ERA5, ERA-Interim, JRA55, MERRA-2) over the Poyang Lake Basin in China, observed improved performance after 2002, highlighting the need for careful temporal evaluation of reanalysis products in hydro-climate analysis.

This study investigates the effectiveness of global climate datasets, specifically AgERA5 (ECMWF) and MERRA-2 (NASA), in the context of small-scale agricultural applications, specifically irrigation management, using as an example the wine-producing region of Nemea, Greece, which is characterized by water scarcity, steep relief, and extensive spatial variability. Even if AgERA5 specifically targets agricultural applications and both datasets are already used in agricultural applications, their effectiveness in small-scale agricultural contexts remains under-explored [27].

Therefore, our primary objective is to assess the reliability of AgERA5 and MERRA-2 as alternatives to local meteorological data, particularly in regions with limited or non-existent weather station coverage. Specifically, we evaluate their performance by comparing irrigation plans derived from these reanalysis datasets with those based on observations from local meteorological stations within representative vineyards. This comparison is conducted in two ways:

(1) Directly comparing weather station data with the corresponding reanalysis grid cells;
(2) Comparing weather station data with downscaled reanalysis data using two spatial interpolation techniques.

This evaluation addresses the critical data scarcity challenges faced by many small-scale agricultural regions.

2. Materials and Methods

2.1. Description of the Study Area

The Nemea Protected Designation of Origin (PDO) wine-region covers an area of about 2700 ha, and it is the largest PDO zone in Greece. It is located in the northeastern part of Peloponnesus, Greece (37°49′4″ N, 22°39′43″ E) (Figure 1a), and holds historical significance for its longstanding viticultural tradition [28]. The region's Mediterranean climate, characterized by warm, dry summers and mild winters with limited precipitation and abundant sunlight, provides optimal conditions for viticulture [15,29]. Nemea's terroir, defined by limestone-rich soils and diverse altitudes ranging from 94 m to 1072 m with an average of 382 m, contributes to the unique flavor profiles of its wines [28]. The steep terrain (Figure 1a) creates a challenging environment with extreme spatial variability in weather and soil conditions, which complicates agricultural water management. For this reason, this area was selected as a good case study for evaluating the usability of global climate datasets for small-scale agricultural applications and, specifically, irrigation management.

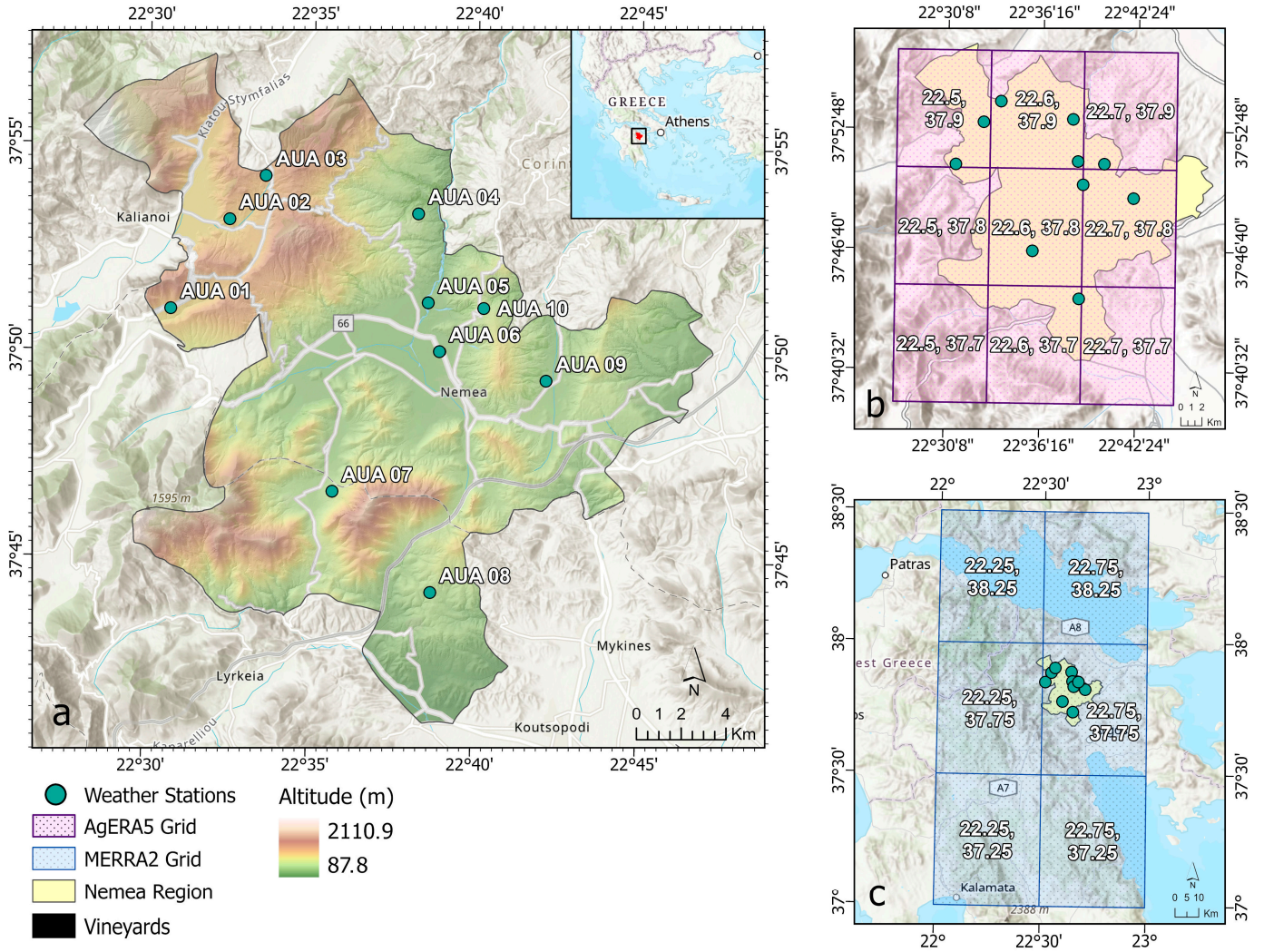


Figure 1. Map of the study area (**a**) where the grids of the AgERA5 (**b**) and MERRA-2 (**c**) climate datasets are overlaid over the weather stations.

2.2. Datasets

The calculation of reference evapotranspiration (ET_o) and the calculation of the water balance in the root zone to optimize irrigation scheduling require the use of specific atmospheric variables, such as precipitation, air temperature, air relative humidity, wind speed, and solar radiation. Soil characteristics, such as soil texture, soil depth, and soil drainage, are also important. The required meteorological data for the needs of our study were acquired from two sources: (i) a dense weather stations network in the study area; (ii) two reanalysis global climate datasets, the AgERA5, and the MERRA-2. The soil-related data were acquired from the soil map of the Nemea wine producing zone.

Additional global climate datasets such as E-OBS, NCEP/NCAR, and JRA-55 were initially considered. An initial assessment indicated that AgERA5 and the MERRA-2 were the most suitable for this application because they had all the required variables, covered the study area, and had a fine spatial resolution (at least finer than the other relevant datasets). For a chronologically homogenous set of data, the study period was set to four complete years, from 1 January 2020 to 31 December 2023.

2.2.1. Meteorological Stations Data

Meteorological data were systematically acquired by the network of monitoring stations of the Spatiotemporal Observatory of the Wine and Viticultural Potential of PDO

Nemea (https://nemeaopap.aua.gr/ accessed on 10 February 2025). The network includes ten automatic, telemetric agrometeorological stations distributed in representative positions (Figure 1) along the broader area of the PDO Nemea zone covering different elevation zones and microclimatic conditions that were installed in 2019 in the framework of the research project "Spatial-temporal Observatory for the Evaluation of Viticultural and Wine Potential OPAP Nemea" and operate constantly up to now. Each station is outfitted with a data logger programmed to capture measurements from all installed sensors at 15 min intervals. The dataset encompasses critical meteorological parameters, including precipitation, wind speed, wind gust, air relative humidity, air temperature, and solar radiation, spanning a period of 5 years, specifically from August 2019 to date. Data retrieval at a quarter-hour resolution was performed from the respective database, subsequently transcribed into Excel format for data quality assessment and further analysis.

Before any further analysis, the retrieved data were assessed for integrity and consistency with semi-automatic supervised data quality testing and a gap-filling algorithm that was developed using MS Excel VBA [30,31]. The algorithm identified gaps and reported obviously erroneous values. To this end, the absolute minimum and absolute maximum recorded values for each parameter were used as thresholds for the identification of erroneous values as well as various other checks such as negative precipitation, sunshine hours, or wind speed values, etc. Marked erroneous values were manually examined in order to understand the source of errors and explore possible easy ways of correcting them in order to avoid the loss of precious data. Finally, the remaining gaps were filled using data from the three nearest stations based on the inverse distance weight (IDW) method with a power of 1 and considering precipitation and temperature lapse rates for precipitation and temperature data, respectively. As this is a relatively new monitoring network, the total number of missing or erroneous values was less than 1%; accordingly, the gap-filling procedure has a negligible impact on the obtained results.

2.2.2. AgERA5

The AgERA5 dataset is a relatively new climate dataset specifically designed for agricultural applications [19]. AgERA5 provides a comprehensive repository of daily surface meteorological information spanning from 1979 to the present day. Derived from hourly ECMWF ERA5 data at the surface level, this dataset has been specifically tailored for the needs of agricultural and agro-ecological investigations. Its primary aim is to streamline the process of acquiring and preprocessing ERA5 data, facilitating practical applications, meaningful analyses, and modeling works. AgERA5 encompasses a wide array of variables that are particularly selected to align with the input requirements of common agricultural and agro-ecological models. Notably, the dataset aggregates data into daily time steps, seamlessly adjusts to local time zones, and rectifies spatial discrepancies to a 0.1° spatial resolution through regression equations calibrated against ECMWF's high-resolution atmospheric model. This refinement significantly enhances the dataset's suitability for applications in areas characterized by complicated topography, diverse land-use patterns, and complex land–sea boundaries. Developed under the guidance of the Copernicus Climate Change Service, AgERA5 stands as an invaluable tool for scholars and practitioners engaged in climate-related research and practical applications within the agricultural field [19]. For our study, 5 variables were downloaded from the Copernicus Climate Change Service (C3S) Climate Data Store (CDS) website. The AgERA5 grid cells that match with the Nemea region are between the coordinates 22.5° E, 37.7° N and 22.8° E, 37.9° N (Figure 1b) and include the following variables: 10 m wind speed, 2 m relative humidity, 2 m temperature, precipitation flux, and solar radiation flux.

2.2.3. MERRA-2

MERRA-2 was developed by the NASA Global Modeling and Assimilation Office (GMAO) using the Goddard Earth Observing System Model (GEOS) version 5.12.4 [32]. It represents the most recent iteration of global atmospheric reanalysis during the satellite era. This dataset spans from 1980 to the present day, typically exhibiting a latency of approximately three weeks post the conclusion of each month.

Acquisition of the data was facilitated through the NASA Prediction of Worldwide Energy Resources (POWER) app, which originated in 2003 as an extension of the SSE project. Initially focused on the SSE component, the POWER project expanded to include two additional datasets pertinent to architectural (e.g., Sustainable Buildings) and agricultural (e.g., Agroclimatology) sectors. The project's ongoing objective is to enhance and broaden the focused parameters within each POWER section. MERRA-2 is used in various hydrological, but also agricultural applications [20–22] even if, unlike AgERA5, it was not specifically developed to address agricultural needs.

The variables of precipitation, all sky surface shortwave downward irradiance, 2 m temperature, 2 m relative humidity, and 2 m wind speed were obtained using the Single Point Tool and the precise coordinates corresponding to each Nemea weather station in 1-day intervals. Notably, given the dataset's spatial resolution of $0.5^\circ \times 0.625^\circ$, only one grid pixel was necessary to cover the entire network of weather stations. However, 6 grids were downloaded for methodology purposes (Figure 1c), which will be discussed below.

2.2.4. Soil Data

Detailed soil mapping of the study area was made in the framework of the research project "Spatial-temporal Observatory for the Evaluation of Viticultural and Wine Potential OPAP Nemea". Soils of the mapped area were taxonomically classified into four main categories at the level of reference soil group—RSG [33]. Specifically, 41.4% of the soils were classified in Leptosols (13,407 ha), 39.8% in Cambisols (12,894 ha), 13.9% in Calcisols (4511 ha), and 4.9% in Fluvisols (1608 ha) RSGs (Figure 2).

The vast majority (94.9%) of the area corresponding to Leptosols RSG is characterized as moderate fine textured in the topsoil (0–30 cm) and most of the area (64%) covered by the specific RSG presents shallow soil profiles (<30 cm). All the soils belonging to Leptosols RSG were mapped as very well drained. Cambisols were characterized as very well drained and well drained in 81.7% of the area occupied by them. The upper part of the soil profile (0–30 cm) in Cambisols RSG presents a moderate fine soil texture, and the subsoil (30–60 cm) is characterized as moderate fine or fine textured in 68.7% of their area. Soil depth was recorded as deep (100–150 cm) and very deep (>150 cm) in 48.1% of the area of Cambisols and as moderate deep (60–100 cm) in 35.1% of the corresponding area. Calcisols were recorded as very well and well drained in 75% of the area that occupies, and as moderately well drained in 22.9% of the same area. The soil texture of Calcisols is moderate fine in the surface part of the soil (0–30 cm) and moderate fine to fine in the subsoil (30–60 cm) in 81.2% of their area. Soil depth was mapped as moderate deep (60–100 cm) and deep (100–150 cm) in 87.5% of the area of Calcisols and as very deep (>150 cm) in 12.5% of the corresponding area. Half of the soils of the Fluvisols RSG are characterized as very well or well drained, and 36% of the area of the specific RSG includes soils that are moderately well drained. All soils of the Fluvisols RSG are deep (100–150 cm) and very deep (>150 cm), presenting a fine textured topsoil in 21% of their area and a moderate fine textured topsoil in 71% of the same area. The texture of the subsurface part of the soil profile (30–60 cm) is characterized as moderate fine in 92% of the Fluvisols' area.

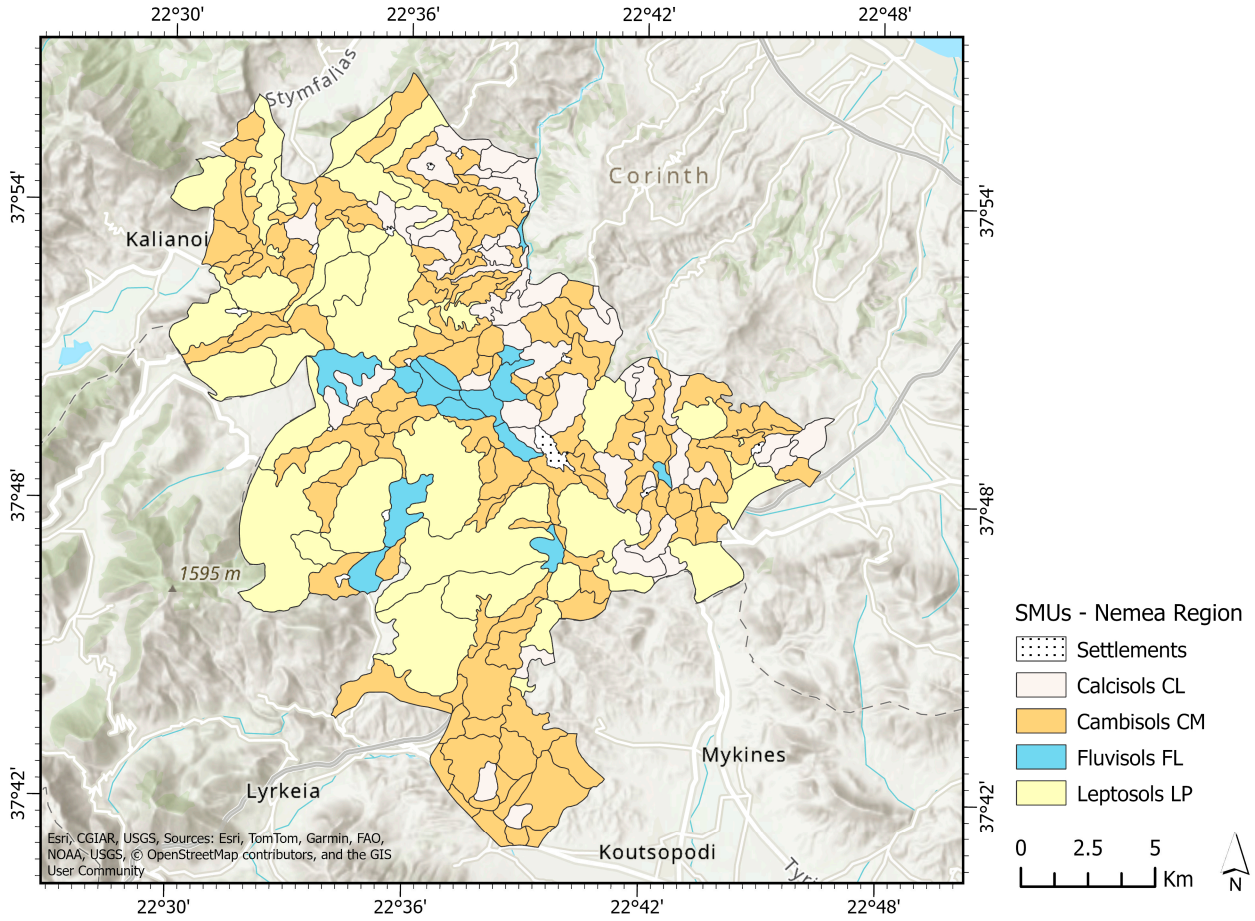


Figure 2. Soil taxonomical classes of the study area at the level of reference soil group.

2.3. Data Pre-Processing

The three datasets are required to be in a 1-day time interval format for them to be compatible with the time step of the applied irrigation management method and to be comparable with each other. Hence, the weather station data and the AgERA5 data were subsequently converted using an MS Excel Visual Basic for Applications (VBA) algorithm. The location of the weather stations was matched with the corresponding grids for the realization of the inter-data comparison.

Additionally, some of the required parameters needed some form of pre-processing for the standardization of their units:

- AgERA5's 2 m temperature (K) was converted to degrees Celsius:

$$T_C = T_K - 273.15 \quad (1)$$

- AgERA5's 10 m wind speed was downscaled to 2 m, using the following equation [34]:

$$u_2 = u_z \frac{4.87}{ln(67.8z - 5.42)} \quad (2)$$

- AgERA5's solar radiation flux ($J/m^2/day$) was converted to solar radiation (W/m^2):

$$Solar\ Radiation(W/m^2) = \frac{Radiation\ Flux(J/m^2/day)}{86,400} \quad (3)$$

- MERRA-2's all sky surface shortwave downward irradiance ($MJ/m^2/day$) was converted to solar radiation (W/m^2):

$$Solar\ radiation(\mathrm{W/m^2}) = \frac{Irradiance(\mathrm{MJ/m^2/day}) \times 1,000,000}{86,400} \quad (4)$$

2.4. Reference Evapotranspiration Calculation

In general, the term evapotranspiration (ET) consists of two phenomena, evaporation through which water is lost from the soil surface and transpiration, where plants evaporate water through canopy. The reference evapotranspiration (ET_o) is the evapotranspiration rate from a reference surface not short of water and represents an indicator of climatic requirements.

Reference evapotranspiration was computed from the meteorological data at a daily time step using FAO Penman–Monteith method, which is considered as the standard method for the definition and computation of the ET_o [34]. The FAO Penman–Monteith method requires radiation, air temperature, air humidity, and wind speed data. An algorithm was developed using VBA in Microsoft Excel to estimate ET_o. This algorithm applies the FAO Penman–Monteith equation on a daily basis, as outlined below:

$$ET_o = \frac{0.408(R_n - G) + \gamma \frac{900}{T+273} u_2 (e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)} \quad (5)$$

where ET_o represents reference evapotranspiration [mm day^{-1}], R_n represents net radiation on the crop surface [MJ m^{-2} day^{-1}], G represents ground heat flux density [MJ m^{-2} day^{-1}], T represents average daily air temperature at 2 m height [°C], u_2 represents wind speed at 2 m height [m s^{-1}], e_s represents saturation vapor pressure [kPa], e_a is actual vapor pressure [kPa], $e_s - e_a$ is saturation vapor pressure deficit [kPa], Δ is vapor pressure curve slope [kPa $°C^{-1}$], and γ represents psychrometric constant [kPa $°C^{-1}$].

This equation constitutes an accurate and straightforward representation of the physical and physiological factors influencing evapotranspiration processes [34].

2.5. Development of a Main Algorithm for Determining Irrigation Needs

The assessment of global datasets' performance in irrigation planning was based on the standard soil water balance method. Soil water balance (SWB) models are theoretical representations of a limited segment of the water cycle, primarily focused on the soil–plant–atmosphere interface [35]. These models hold significance in agricultural water management, as they are widely used for irrigation scheduling, representing the most common method for determining the timing and amount of water required for crop irrigation [36].

The first step was to estimate the required soil hydraulic properties based on the soil characteristics and depth using pedotransfer functions. The required properties are porosity, total available soil water (*TAW*), and readily available soil water (*RAW*). *TAW* refers to the water-holding capacity estimated as the product of the depth of the root zone (in mm) multiplied by the difference between the water content of the soil at field capacity and at the wilting point (both in mm^3/mm^3) [36]. RAW is defined as the maximum depletion of soil water within the root zone for which the actual evapotranspiration ET_a is equal to the crop potential evapotranspiration ET_p [34]. *RAW* is calculated using the following equation [37]:

$$\mathrm{RAW} = \mathrm{p} \times \mathrm{TAW} \quad (6)$$

where p is the average fraction of *TAW* that can be depleted from the root zone before moisture stress (water stress, ET reduction) occurs. The critical depletion factor (p) represents the critical soil moisture level where water stress occurs, which affects evapotranspiration and crop production. Its values range between 0.4 and 0.6, with lower values obtained for

sensitive crops with limited root systems under high evapotranspiration conditions and higher values for deep and dense roots and low evapotranspiration rates. In this study, the p factor is equal to 0.45, according to the literature, since it concerns wine-producing vineyards [34].

Following this, effective precipitation was determined with a simplified approach as 80% of the total daily precipitation. This simplified approach provides a reasonable approximation in dry areas [38,39] where precipitation has an inferior role in irrigation requirements, while any deviations will be the same for examined scenarios and will not influence the comparison. Having calculated the reference evapotranspiration, the next step was to estimate the ET_p. The latter differs distinctly from the ET_o as the ground cover, canopy properties, and aerodynamic resistance of the actual crop (vines) are different from the reference crop (grass). Potential evapotranspiration was calculated as the product of the crop coefficient (K_c) and ET_o, represented by the equation $ET_p = K_c \times ET_o$.

Crop coefficients (K_c) are generally empirical ratios of potential evapotranspiration to reference evapotranspiration and are derived from experimental data. A common formula for the crop coefficient is $K_c = ET_p/ET_o$, where K_c is the dimensionless crop coefficient for a particular crop at a given growth stage [34].

In the context of this paper, the K_c values were derived from the literature. According to Food and Agriculture Organization of the United Nations (FAO) Irrigation and Drainage paper no. 56, the K_c values for grapes cultivated for wine production are 0.3 in the initial growth stage, 0.7 in the mid-growth stage, and 0.45 in the late-season stage. The same K_c values were used in all cases to provide comparable results and allow the evaluation of the global datasets [34].

Utilizing the root zone water balance (SWB), the water depletion (soil water depletion—D) was computed on a daily basis, facilitating the estimation of the soil water stress coefficient (K_s). The SWB model was formulated to track changes in water stored within the root zone control volume, denoted by the equation:

$$\mathrm{D_{r,i}} = \mathrm{D_{r,i-1}} - (\mathrm{P_{p,i}} - \mathrm{RO_i})_\mathrm{i} - \mathrm{CR_i} - \mathrm{I_i} + \mathrm{ET_{a,i}} + \mathrm{DP_i} \quad (7)$$

where $D_{r,i}$ represents soil moisture depletion on a given day, $D_{r,i-1}$ represents depletion on the previous day, $ET_{a,i}$ denotes the actual evapotranspiration, $P_{p,i}$ denotes precipitation, RO_i represents runoff (and their difference reproduces effective precipitation), CR_i represents capillary rise, I_i represents irrigation, and DP_i represents deep percolation on day i. Capillary rise from the groundwater table was deemed negligible since the aquifer in the area is at a great depth.

In this study, the water stress coefficient ($\mathrm{K_s}$) was determined using the following equation,

$$\begin{gathered} K_{s,i} = 1 \text{ if } D_{r,i} \le RAW \\ K_{s,i} = \frac{TAW - Dr,i}{TAW - RAW} \text{ if } D_{r,i} \ge RAW \end{gathered} \quad (8)$$

where TAW represents the total available water in the root zone (mm), RAW indicates the percentage of TAW utilized by the canopy without affecting transpiration, and $\mathrm{D_{r,i}}$ signifies the daily water depletion derived from the soil water balance.

If the water deficit is equal to or lower than the readily available water, K_s is assigned a value of 1; otherwise, it is determined as $(TAW - D)/(TAW - RAW)$ when the water depletion surpasses the value of readily available water.

Simultaneously, the ET_a is equal to the product of the K_s and the ET_p, $ET_a = K_s \times ET_p$. Furthermore, the available soil water obtained by subtracting the water depletion of the current day from the total available soil water $(TAW - D)$, was also assessed.

Lastly, irrigation requirements were determined, indicating the necessary daily water application to meet crop needs. As per the algorithm, if the current day's water depletion fell below the readily available soil water, irrigation was unnecessary. However, if the depletion exceeded the *RAW* value, an amount equivalent to the water depletion on that day was applied.

2.6. Spatial Interpolation Methodology

Assuming the geometric center of each pixel as an individual point, the data were downscaled using three different variations of inverse distance weighting (IDW), one simple IDW (Equation (9)) and two IDW methods enhanced by the inclusion of temperature–elevation and precipitation–elevation coefficients (Equation (10))

$$x_p = \frac{\sum_{i=1}^{n} \frac{x_i}{d_i}}{\sum_{i=1}^{n} \frac{1}{d_i}} \tag{9}$$

$$x_p = \frac{\sum_{i=1}^{n} \left[\frac{(h_p - h_i) \times c}{d_i} + \frac{x_i}{d_i} \right]}{\sum_{i=1}^{n} \frac{1}{d_i}} \tag{10}$$

where x_p is the interpolated value of point p, x_i is the observed value of the i-th neighbor, d_i is the distance between the point p and the i-th neighbor, n is the number of neighbors, h_p is the elevation of point, h_i is the elevation of the i-th neighbor, c is the coefficient value.

These coefficients were calculated daily for temperature and precipitation based on a regression between the elevation of pixels centers and the precipitation and temperature values of AgERA5 and MERRA-2, respectively, for each day. Then, one of the two methods used the appropriate coefficient value for each respective day, while the other used a single coefficient value calculated by averaging the daily values. For the rest of the variables, simple IDW was used in every case as no significant correlation with elevation was discovered. It should be noted that to provide a representative evaluation of the global datasets, the study area was considered ungauged; therefore, the rainfall and temperature lapse rates were calculated utilizing only the AgERA5 and MERRA-2 and not the observed data from the 10 meteorological stations.

2.7. Reanalysis Models Evaluation Methodology

The validation of the AgERA5 (ECMWF) and MERRA-2 (NASA) datasets against local meteorological station data involved several key steps. The analysis focused on two critical meteorological variables for irrigation management, i.e., precipitation and ET_p and the resulting irrigation water requirements. The comparisons were made, including between the observed data and the nearest pixel of the metanalysis datasets ("AgERA5 Raw" and "MERRA-2 Raw"), the IDW without elevation compensation ("AgERA5 IDW Simple" and "MERRA-2 IDW Simple"), the IDW with average lapse rates ("AgERA5 IDW Average Coefficient" and "MERRA-2 IDW Average Coefficient"), and the IDW with the daily lapse rates ("AgERA5 IDW Daily Coefficient" and "MERRA-2 IDW Daily Coefficient"). The following metrics were used in the evaluation:

- Regression lines (intercept, slope, and coefficient of determination R^2)

As a first step in the analysis, the estimated values based on AgERA5 and MERRA-2 datasets were plotted against the corresponding values based on the observed data. Then, linear regression lines were used to assess the strength of the relationship between the estimated and the observed values. The primary metric used in this case to evaluate the models' performance was the coefficient of determination (R^2) with values closer to 1 meaning better performance and values lower than 0.5 indicating a weak correlation.

Furthermore, an intercept of the linear regression near 0 and a slope near 1 indicate better performance (estimated values coinciding with the observed values).

- Root Mean Square Error

The root mean square error (RMSE) is a widely used metric to assess the accuracy of a model's predictions. It measures the average magnitude of the errors between predicted and observed values, giving greater weight to larger errors by squaring them before averaging. RMSE is expressed in the same units as the data, making it easy to interpret. A lower RMSE indicates a better fit, while a higher RMSE suggests that the model's predictions deviate significantly from the actual values. It is particularly useful for comparing different models or algorithms to evaluate their predictive performance [40].

$$RMSE = \sqrt[2]{\frac{\sum_{i=1}^{n}\left(y_i - y_{pi}\right)^2}{n}} \quad (11)$$

where y_i is the observed value of the i-th time step, y_{pi} is the predicted value of the i-th timestep, n is the number of timesteps.

- Mean Bias Error

The mean bias error (MBE) is a metric used to evaluate the bias in models' predictions. It calculates the average difference between predicted and observed values, indicating whether a model systematically over- or underestimates the actual values. A positive MBE suggests that the model tends to underestimate, while a negative MBE indicates overestimation. Unlike other metrics, MBE focuses on the direction of errors rather than their magnitude [41].

$$MBE = \frac{\sum_{i=1}^{n}\left(y_i - y_{pi}\right)}{\overline{y}} \quad (12)$$

where y_i is the observed value of the i-th time step, y_{pi} is the predicted value of the i-th timestep, n is the number of time steps, and $\overline{y}$ is the average of all observed values.

Finally, the significance of the differences between the irrigation water requirements calculated using the observed data and using two reanalysis datasets analysis was evaluated with ANOVA test.

The creation of plots were assisted by R language (R version 4.4.1) code and especially the dplyr and ggplot2 packages that were used to create the graphs [42–44].

3. Results

3.1. Reanalysis Models Validation

3.1.1. Precipitation Analysis

The analysis of precipitation data highlights the very large differences in the performance of AgERA5 and MERRA-2 datasets, particularly when varying spatial interpolation methods are applied for downscaling the datasets. For AgERA5, the raw data yielded an R^2 of 0.10, indicating a very weak correlation with the observed precipitation. Application of simple IDW improved the correlation slightly, with an R^2 of 0.22. Further refinement using IDW with daily elevation corrections provided a marginal improvement, while the IDW incorporating the daily average elevation corrections provided similar results. Specific details on correlation coefficients and the slopes and intercepts of the fitted lines are presented in Figure 3, and a visual inspection of the relevant scatterplots provides a similar picture.

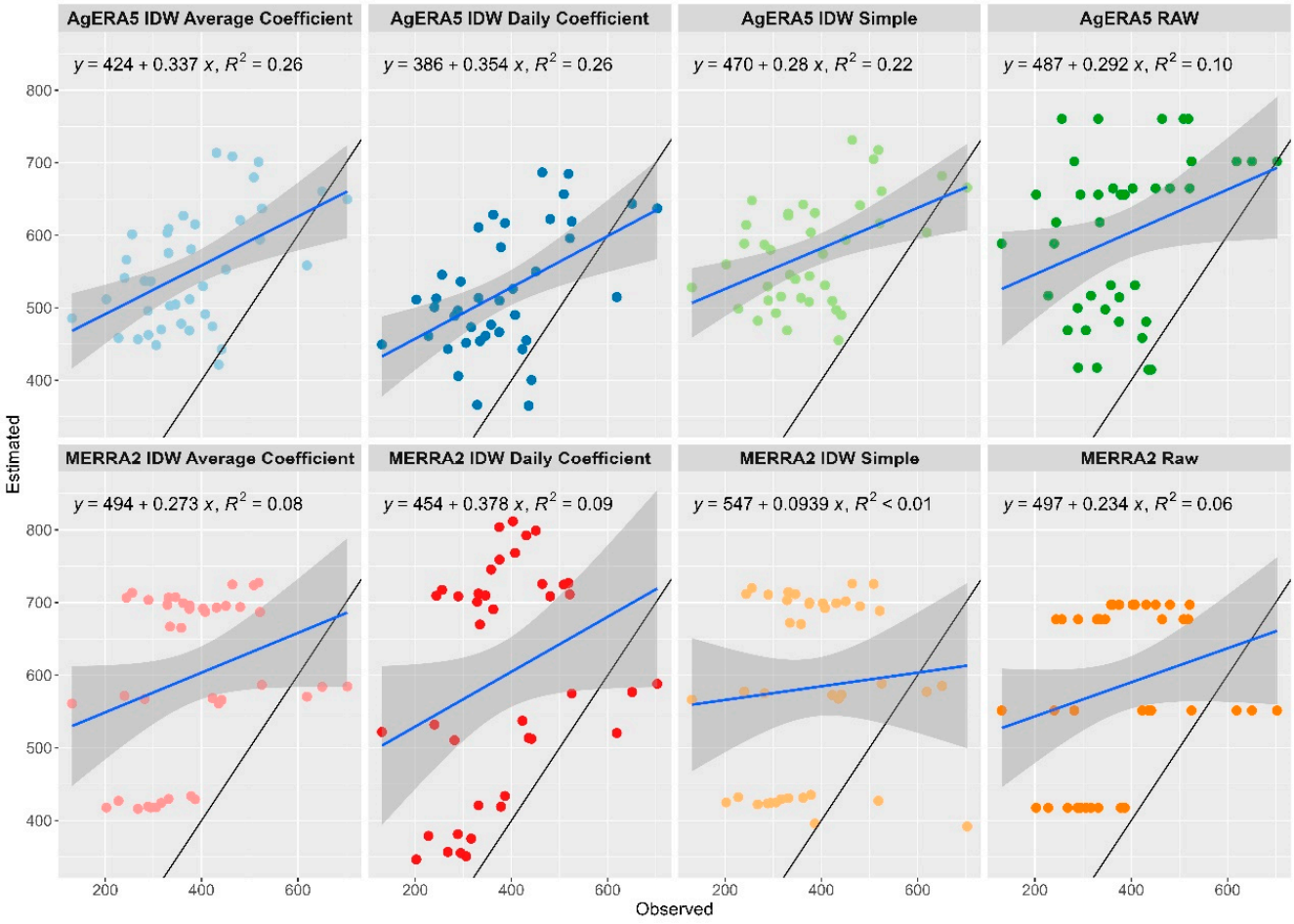


Figure 3. Relationship between observed and estimated total annual precipitation values using AgERA5 and MERRA-2 datasets with various interpolation methods. The linear regression lines in blue color, the 95% confidence intervals, and the x = y lines in black color are also plotted.

The MERRA-2 dataset exhibited even lower correlations consistently. The raw data produced an R^2 of 0.06, with marginal or no improvements observed when applying IDW methods. In simple IDW the R^2 was less than 0.01, while IDW with daily and average elevation corrections provided moderate improvements, with R^2 values of 0.09 and 0.08, respectively. These findings suggest that while AgERA5 consistently outperforms MERRA-2 in predicting precipitation, both datasets struggle to fully capture the localized variability in the study area, as is indicated by the very low slope values in all cases. The lower slope values in MERRA-2 (see Figure 3) can be also attributed to the lower spatial resolution of this dataset. The very high intercept values highlight the increased bias in lower precipitation values.

Table 1 presents the average precipitation values for the 4-year period studied across the 10 locations. It is clear that both datasets generally overestimate precipitation compared to the observed values. AgERA5 predictions are consistently higher, especially for the raw data, while MERRA-2 shows slightly lower estimates. The interpolation methods (simple IDW, daily coefficient, and average coefficient) improve the accuracy of the predictions by reducing the overestimation, with the IDW average coefficient method providing the closest estimates to the observed precipitation values across the 10 stations.

Table 1. Average total annual precipitation values of the 4-year period for each station.

		AgERA5				MERRA-2			
Station	**Precipitation Observed**	**Raw**	**IDW Simple**	**IDW Daily Coef**	**IDW Avg Coef**	**Raw**	**IDW Simple**	**IDW Daily Coef**	**IDW Avg Coef**
					(mm)				
1	483.3	695.8	646.4	613.7	622.8	585.7	609.5	607.5	607.8
2	481.6	695.8	671.1	640.8	649.7	585.7	609.2	607.9	608.1
3	492.5	695.8	664.1	641.7	648.2	585.7	411.5	609.8	610.0
4	404.9	695.8	606.3	536.6	562.2	585.7	605.8	598.0	599.3
5	304.6	695.8	587.0	509.8	538.4	585.7	603.3	595.2	595.9
6	352.9	445.3	500.1	431.0	455.6	585.7	601.7	593.9	594.7
7	301.9	563.4	562.2	494.2	517.1	585.7	602.6	596.1	597.4
8	262.8	563.4	521.3	460.3	481.2	585.7	585.2	578.9	580.0
9	365.9	445.3	475.7	407.2	548.8	585.7	598.0	590.9	591.7
10	358.1	492.4	530.5	477.5	496.5	585.7	601.9	596.8	597.8

3.1.2. Potential Evapotranspiration (ET_p) Analysis

The comparison of ET_p values (Figure 4) reveals similar trends, with AgERA5 generally performing somehow better or similar to MERRA-2. For AgERA5, the raw data resulted in an R^2 less than 0.01, demonstrating minimal correlation with the observed ET_p values. The use of simple IDW slightly improved the correlation, with further gains achieved through IDW with daily elevation correction. The most obvious improvement came from the application of IDW with average elevation correction, which produced an R^2 of 0.03. In all cases, however, the correlations were very weak.

MERRA-2's performance shows similar results. The raw data yielded an R^2 of 0.05, and the application of simple IDW produced a slight decrease in correlation, with R^2 value of 0.01. The use of daily elevation correction, and the average elevation correction could not improve the performance. These results emphasize the limited utility of both AgERA5 and MERRA-2 for potential evapotranspiration predictions in this region. In all cases, the slope was close to 0, indicating that both datasets cannot capture the spatial variability in the study area (the observed values vary obviously in small distances while both datasets provide very similar values for the entire area). These observations and all the statistics are visualized in Figure 4.

Table 2 provides the average annual potential evapotranspiration values over the 4-year period for the 10 stations. Both datasets tend to overestimate evapotranspiration, with AgERA5 showing higher deviations, particularly in the raw data. The interpolation methods enhance prediction accuracy by reducing this overestimation. Notably, the IDW average coefficient method yields estimates that, on average, align more closely with the observed values, demonstrating improved performance for most stations in both datasets.

Table 2. Average evapotranspiration values of the 4-year-period for each station.

		AgERA5				MERRA-2			
Station	**ET_p Observed**	**Raw**	**IDW Simple**	**IDW Daily Coef**	**IDW Avg Coef**	**Raw**	**IDW Simple**	**IDW Daily Coef**	**IDW Avg Coef**
					(mm)				
1	561.4	613.6	625.3	592.4	593.3	548.4	688.7	653.5	686.5
2	602.5	613.7	625.9	594.2	595.1	548.4	689.3	654.0	688.0

Table 2. *Cont.*

Station	ET_p Observed	AgERA5 Raw	AgERA5 IDW Simple	AgERA5 IDW Daily Coef	AgERA5 IDW Avg Coef	MERRA-2 Raw	MERRA-2 IDW Simple	MERRA-2 IDW Daily Coef	MERRA-2 IDW Avg Coef
3	503.2	659.6	625.2	613.7	614.3	548.4	690.3	655.0	691.0
4	493.6	659.6	645.7	609.7	611.6	548.4	687.8	651.9	678.5
5	524.0	659.8	648.1	608.2	610.3	548.5	688.8	652.5	677.8
6	542.4	705.2	663.0	625.1	627.0	548.5	690.1	653.7	679.8
7	546.1	682.3	647.4	611.7	613.5	548.4	690.5	654.3	683.0
8	596.6	701.0	674.6	641.5	643.2	548.4	697.7	660.6	690.3
9	563.7	705.0	667.4	627.6	621.1	548.4	692.8	656.0	683.6
10	604.0	700.5	660.7	632.3	633.8	548.4	691.8	655.5	685.9

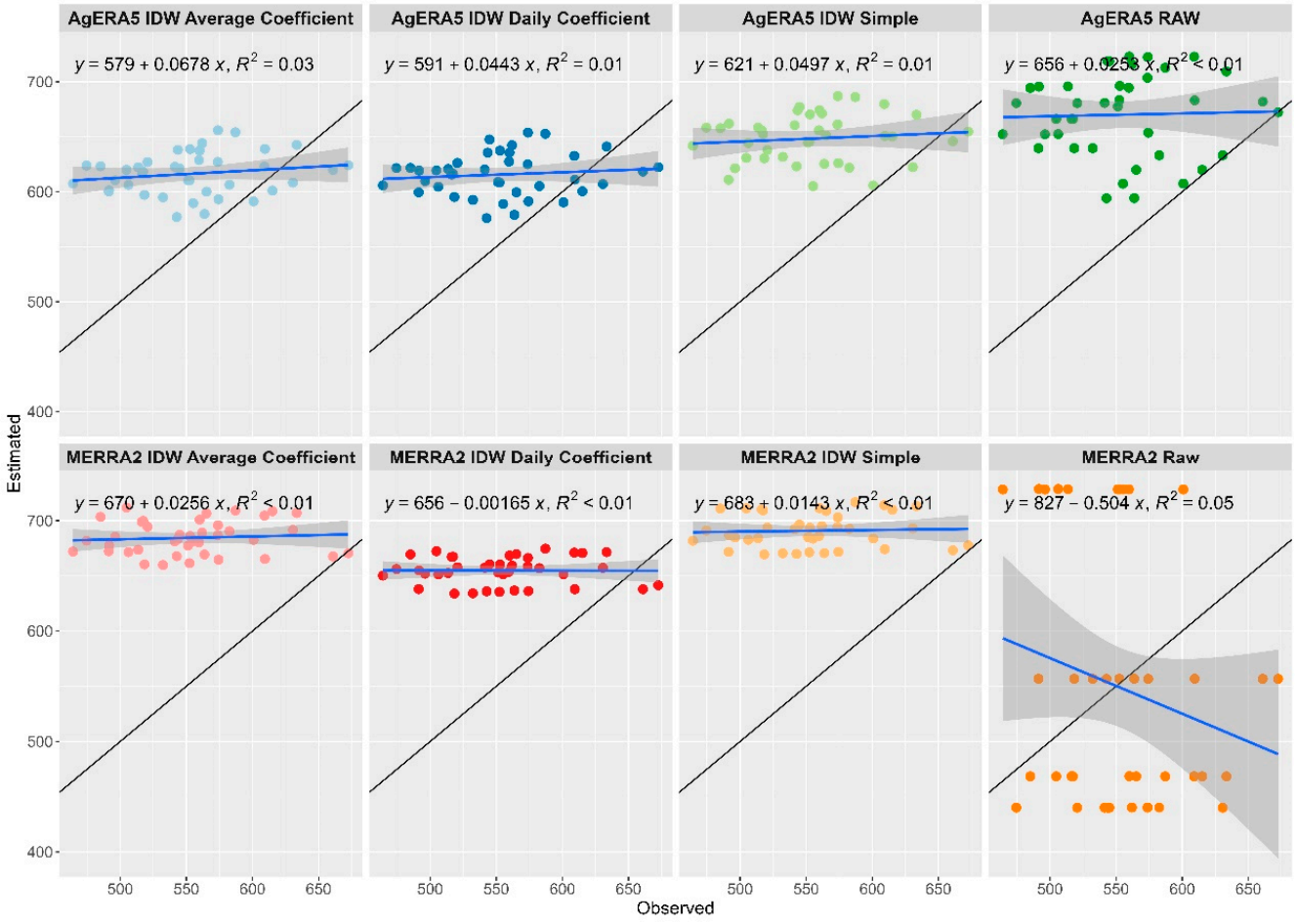


Figure 4. Relationship between observed and estimated total annual potential evapotranspiration values using AgERA5 and MERRA-2 datasets with various interpolation methods. The linear regression lines in blue color, the 95% confidence intervals, and the x = y lines in black color are also plotted.

3.1.3. Irrigation Planning Analysis

The results for irrigation planning show a mixed performance between AgERA5 and MERRA-2. For AgERA5, the raw data produced an R^2 of 0.12, which improved slightly with simple IDW. The IDW with daily elevation correction provided a comparable correlation, while IDW with average elevation correction slightly reduced the correlation, yielding an R^2 of 0.11.

Interestingly, MERRA-2 performed better than AgERA5 in some instances for irrigation planning. The raw data produced a lower correlation, but the application of simple IDW significantly increased the correlation to $R^2 = 0.22$. The IDW with average elevation correction provided the highest correlation for MERRA-2, with an R^2 of 0.23, while daily elevation correction yielded a slightly lower R^2 of 0.17 but a slope closer to 1. These findings suggest that MERRA-2 may be more suitable than AgERA5 for certain aspects of irrigation planning, though both datasets have poor performance. Specific details on correlation coefficients and the slopes and intercepts of the fitted lines are presented in Figure 5.

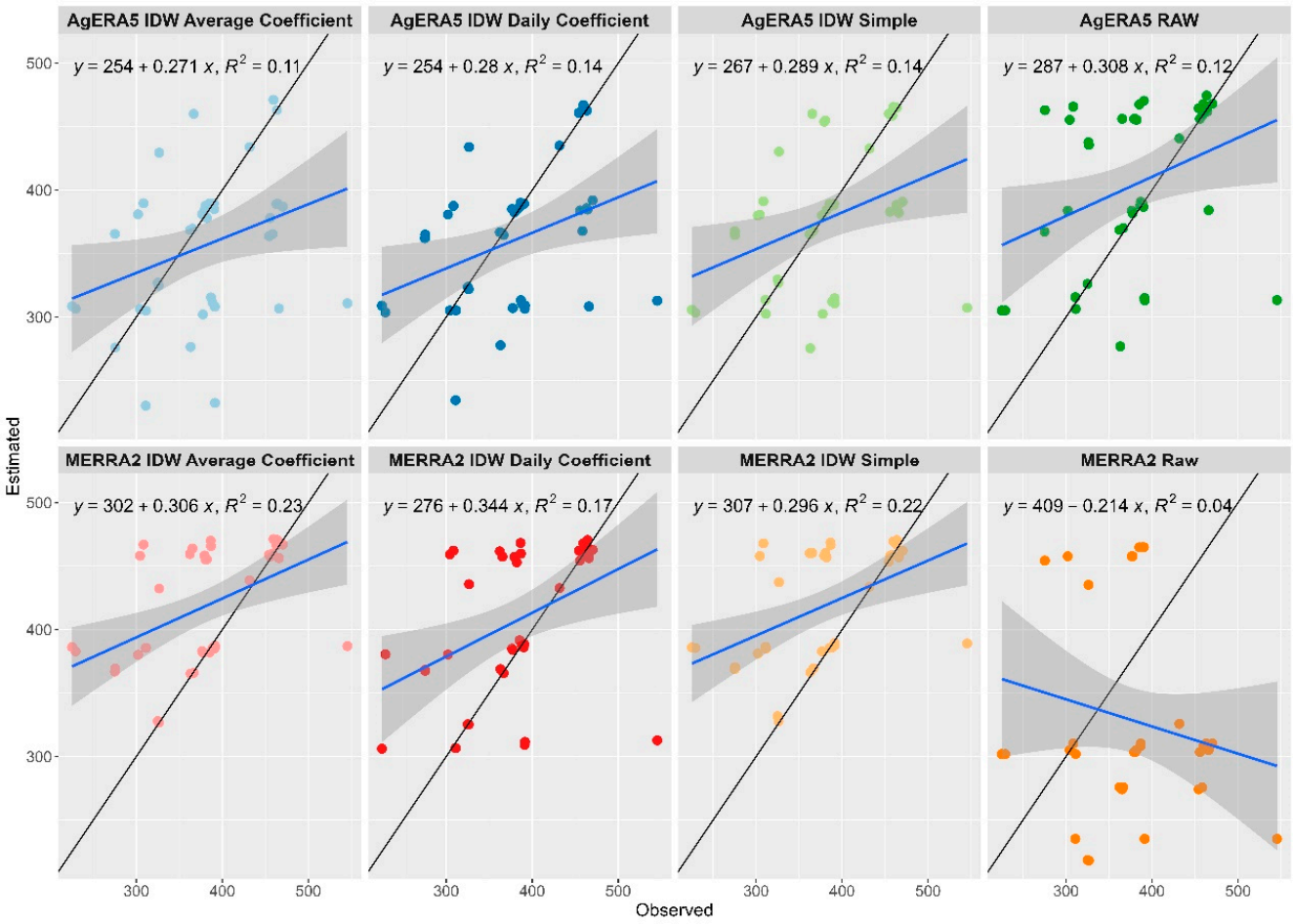


Figure 5. Relationship between annual irrigation requirements values calculated using observed vs. estimated data using AgERA5 and MERRA-2 datasets with various interpolation methods. The linear regression lines in blue color, the 95% confidence intervals, and the x = y lines in black color are also plotted.

Table 3 displays the average annual irrigation requirements for the 10 stations over the 4-year period. Again, while AgERA5 raw data tend to slightly overestimate irrigation for most stations, the predictions are generally closer to the observed values compared to MERRA-2, which shows larger overestimations, particularly in the raw and IDW simple methods. Accordingly, the interpolation techniques help to reduce discrepancies in both datasets. Notably, the IDW average coefficient method provides the most accurate estimates for AgERA5 across several stations, whereas MERRA-2 performs better with the IDW daily coefficient approach for some locations.

Table 3. Average annual irrigation requirements of the 4-year period for each station.

		AgERA5				MERRA-2			
Station	**Irrigation Observed**	**Raw**	**IDW Simple**	**IDW Daily Coef**	**IDW Avg Coef**	**Raw**	**IDW Simple**	**IDW Daily Coef**	**IDW Avg Coef**
					(mm)				
1	368.2	371.2	350.2	312.0	311.0	329.6	427.0	405.5	426.6
2	402.5	382.2	342.6	326.2	323.0	342.1	420.7	400.4	420.4
3	321.3	400.1	380.1	345.4	345.0	342.1	421.3	402.0	421.5
4	323.5	400.1	380.7	363.4	363.4	342.1	421.1	417.6	418.5
5	341.5	367.2	367.1	343.2	345.1	320.1	414.0	414.1	414.0
6	352.4	410.1	379.9	378.7	378.8	299.5	382.7	379.8	381.5
7	388.6	406.9	369.3	367.7	350.2	329.6	428.2	407.6	427.4
8	464.3	429.4	387.4	387.0	388.1	329.6	428.1	408.2	426.5
9	388.7	439.2	412.9	389.6	366.3	320.1	413.2	414.1	413.5
10	427.9	430.4	390.2	388.7	387.9	329.6	427.3	407.2	427.7

Interestingly, while the difference in the annual irrigation requirements is much higher, as can be seen in Figure 5 and as can be indicated by the high RMSE values (Table 4), the average annual irrigation requirements estimated with the metanalysis datasets are closer to the observed ones, indicating that the differences are balanced through the years.

Table 4. Root mean square error (RMSE) of the two datasets for the three studied variables.

	AgERA5				MERRA-2			
Variable	**Raw**	**IDW Simple**	**IDW Daily Coef**	**IDW Avg Coef**	**Raw**	**IDW Simple**	**IDW Daily Coef**	**IDW Avg Coef**
				RMSE (mm)				
Precipitation	220.5	198.2	154.9	177.6	220.7	227.5	231.0	229.1
ET_p	116.3	96.6	69.0	69.3	112.3	137.0	103.7	130.8
Irrigation	54.2	49.8	50.0	54.9	111.5	49.8	49.4	49.7
				ANOVA for irrigation water requirements				
ANOVA F/*p*-value	21.8/ 0.000	14.7/ 0.000	13.3/ 0.000	28.6/ 0.000	21.8/ 0.000	14.7/ 0.000	13.3/ 0.000	28.6/ 0.000

3.2. Predictive Accuracy and Bias Assessment

Table 4 presents the RMSE values for precipitation, ET_p, and irrigation across various interpolation methods for both AgERA5 and MERRA-2 datasets. Lower RMSE values indicate better predictive accuracy.

- Precipitation: For AgERA5, the RMSE decreases significantly when using interpolation methods compared to the raw data, with the best performance being a 29.8% decrease in RMSE seen in the IDW daily coefficient (154.9 mm). In contrast, MERRA-2 has generally higher RMSE values, with the raw data at 220.7 mm and interpolation methods yielding marginal deteriorations, the worst being an RMSE of 231 mm, a 9.3% increase in RMSE, with IDW daily coefficient. A general observation is that RMSE values are very high considering the low precipitation depths in the area.
- Potential evapotranspiration (ET_p): AgERA5 again shows a clear improvement with interpolation, particularly the IDW daily coefficient, which reduces the RMSE from 116.3 mm (raw) to 69 mm, signifying a 46.3% decrease in RMSE, with the IDW average

coefficient being a close second (69.3 mm). MERRA-2 exhibits marginally better performance in the raw data with RMSE = 112.3 mm, and higher errors with interpolation methods (137 mm or a 22% increase in IDW simple and 130.8 mm or a 16.5% increase in IDW average coefficient), with the exception of IDW daily coefficient (103.7 mm), which exhibits a 7.7% decrease in RMSE.

- Irrigation: The RMSE for AgERA5 shows minimal variation between the raw data and various interpolation methods, with values ranging from 49.8 mm to 54.9 mm, a relative difference of −8.1% to +1.3%. For MERRA-2, the raw data results in a higher RMSE of 111.5 mm, though interpolation reduces the error significantly, with the IDW daily coefficient producing the lowest RMSE of 49.4 mm, which constitutes a 56% reduction in RMSE and marks the best performing dataset of all. However, the high RMSE values in all cases highlight the high risks of over-irrigation or under-irrigation when using these datasets, suggesting that these datasets may have limited utility for applications requiring high precision irrigation management.

Overall, the RMSE results show that AgERA5 consistently performs better than MERRA-2, particularly for precipitation and irrigation, and especially when interpolation methods are applied; however, it has a slightly better RMSE value for the case of evapotranspiration. Comparing the best RMSE values obtained for AgERA5 and MERRA-2, it is observed that AgERA5 has 42% lower RMSE in precipitation and 5% lower RMSE in potential evapotranspiration, while the best RMSE in irrigation water requirements is about the same for both datasets.

As can be also seen in Table 4, where the analysis of variance F-ratio and significance values for the irrigation estimates are presented, the differences between the irrigation water requirements produced by the observed data and the evaluated datasets are significant in all cases.

Table 5 presents the MBE values, which reflect the overall bias in the model predictions. Positive MBE values indicate underestimation, while negative values indicate overestimation.

Table 5. Mean bias error (MBE) of the two datasets for the three studied variables.

	AgERA5				MERRA-2			
Variable	**Raw**	**IDW Simple**	**IDW Daily Coef**	**IDW Avg Coef**	**Raw**	**IDW Simple**	**IDW Daily Coef**	**IDW Avg Coef**
				MBE (mm)				
Precipitation	−218	−196	−140	−171	13	−202	−217	−217
ETp	−116	−95	−62	−63	122	−137	−101	−131
Irrigation	−26	2	18	22	75	−40	−28	−40

- Precipitation: For AgERA5, the MBE decreases with interpolation, with IDW daily coef. showing the lowest bias (140 mm), compared to 218.0 mm in the raw data. However, there is overestimation in all cases. MERRA-2 has a very low underestimation for raw but very big overestimation for all other cases.
- Potential evapotranspiration (ET_p): AgERA5 shows a notable reduction in bias with interpolation, but there is overestimation in all cases. In MERRA-2, the raw data have a positive bias, but interpolation methods lead to substantial overestimation, with MBE values reaching −137 mm.
- Irrigation: AgERA5 shows minimal bias with interpolation methods, particularly with IDW simple. MERRA-2 shows a significant underestimation in the raw data, but interpolation leads to big overestimation.

Overall, the MBE analysis indicates that AgERA5 demonstrates reduced bias by applying spatial interpolation methods, particularly for evapotranspiration and irrigation. On the other hand, MERRA-2 shows higher and more variable bias across all parameters, especially for precipitation and evapotranspiration, where interpolation leads to substantial overestimation. It should be noted that MBE measures the average tendency to underestimate or overestimate. A low MBE can be misleading; if underestimation and overestimation are large but fluctuate between years and stations, the errors can largely cancel out, resulting in a low MBE that masks significant inconsistencies.

4. Discussion

AgERA5 is specifically designed with agricultural applications in mind. Furthermore, both AgER5 and MERRA-2 are already extensively utilized in agricultural applications, especially in areas lacking measured meteorological data [20–22,24,25]. Our study found that AgERA5 and MERRA-2 have serious limitations when used for irrigation scheduling in regions characterized by vast spatial variability and data scarcity, such as the Nemea region of Greece, a viticultural area characterized by complex terrain. These results indicate that it is of imperative importance to assess the global datasets in local contexts before proceeding with their operational use in irrigation planning.

Previous studies in other regions indicate that there is generally good agreement with real data [23–25]. However, these studies concerned different areas with different characteristics. Several factors may explain the limited performance of both datasets. The low spatial resolution of these datasets compared to the steep relief and diverse microclimates in the study area is probably the most significant limitation. This limitation is also indicated by the lower effectiveness of MERRA-2, probably due to its lower spatial resolution, which significantly limits its effectiveness in capturing the localized variations characteristic of the study region.

Previous research has also identified similar limitations with the accuracy of reanalysis datasets, particularly in regions with complex terrain or limited ground observations. For example, Li et al. [10] found that the accuracy of reanalysis precipitation datasets varied significantly over time and across different regions in China, emphasizing the importance of a careful evaluation considering the specific contexts. Other studies indicate that while reanalysis datasets are valuable tools, their spatial resolution can be a limiting factor in regions with high spatial variability [45,46].

Despite the aforementioned limitations, AgERA5 generally outperforms MERRA-2 in accuracy and bias, particularly when spatial interpolation is used for downscaling. While less accurate overall, MERRA-2, when enhanced with interpolation, shows potential for irrigation planning. This apparent accuracy stems from a compensation of errors. MERRA-2 tends to overestimate both precipitation and evapotranspiration, leading to a cancellation of errors in the calculated irrigation requirements. This suggests the performance may be coincidental and unreliable. Moreover, MERRA-2's coarser spatial resolution limits its ability to capture the fine-scale variations typical of the study area, a common challenge in regions with complex terrain.

Reanalysis datasets are generated by assimilating various observational data sources, including satellite data, ground weather station data, and other monitoring data, into numerical weather prediction models [9]. The lack of sufficient observation data for reanalysis data calibration in Greece could contribute to the lower performance compared with other studies. As can be seen in Figure 6, the observation points density in Greece is very low and generally much lower than other European countries. Other studies concluded that reanalysis products may yield varying results due to differences in their input data sources [7,11]. This data scarcity issue is prevalent in other countries as well,

further emphasizing the need for accurate local data and regional evaluation of global datasets. An additional problem could be the existence of many gaps in the historical meteorological data series in Greece [16,31].

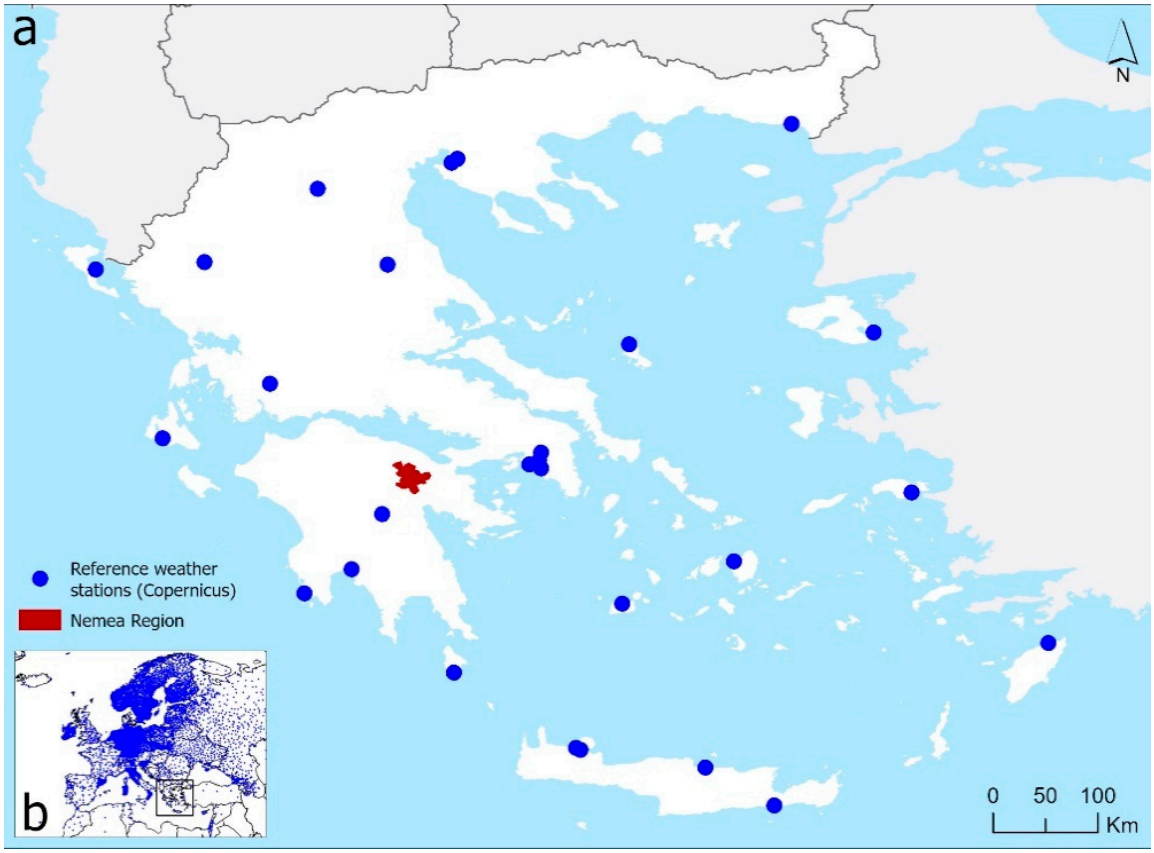


Figure 6. Map with the station coverage which is the basis for the E-OBS precipitation dataset for (**a**) Greece and (**b**) all over Europe. E-OBS is a land-only gridded daily observational dataset for precipitation, temperature, sea level pressure, global radiation, wind speed, and relative humidity in Europe [47].

Our results indicate that using these datasets for irrigation management carries potential risks. Over-irrigation can lead to reduced product quality and potential disease outbreaks, while under-irrigation may result in decreased yields. Mirás-Avalos et al. [48] state that over-irrigation in vineyards can lead to increased susceptibility to fungal diseases, reduced fruit quality, and excessive vegetative growth. Romero et al. [49] also found that over-irrigation in vineyards can result in lower grape quality and increased water consumption. In contrast, Campos et al. [36] found that under-irrigation in vineyards can lead to decreased yields and reduced fruit quality. Lorite et al. [26] demonstrated that proper irrigation scheduling, avoiding under-irrigation, is crucial for maximizing crop yields in semi-arid regions. Considering the high obtained RMSE values, the wasted water in the cases in which we have over-irrigation can be important. In the context of water scarcity, which is a growing concern in many viticultural regions, the accurate estimation of irrigation needs becomes even more critical. Over-irrigation can strain limited water resources, while under-irrigation can lead to economic losses for farmers [49–51]. Therefore, the selection of appropriate datasets and downscaling techniques is crucial for sustainable water management in agriculture.

An interesting finding of the current study was that the observed errors were generally higher in precipitation and lower in ET_p estimation. This resulted in generally lower errors in irrigation requirements estimation, which is reasonable as ET_p has a more crucial role in

irrigation planning because the growing season in the study area coincides with the dry period. Further, as both examined datasets generally overestimate both precipitation (inflow) and evapotranspiration (outflow), the combined errors are smaller. Vanella et al. [25] also reported a good agreement for variables such as air temperature and relative humidity, with ERA5-Land showing slightly higher accuracy for estimating reference evapotranspiration.

Our study did not include calibration or bias correction because we aimed to simulate a scenario where observed data are unavailable, representing the conditions under which these datasets would be most valuable. If observed data existed, direct calibration would be possible, potentially improving the accuracy of the datasets. However, in many data-scarce regions, such calibration may not be feasible, highlighting the importance of evaluating the raw performance of these datasets.

While this study provides valuable insights, it is crucial to acknowledge its limitations. The analysis focused on a specific crop (vineyards), region (Nemea, Greece), and time period (four years), limiting the generalizability to other crops, regions (particularly those with complex terrain), or longer time scales. Future research should address this by investigating the performance of these datasets in diverse agricultural settings and over extended periods. Additionally, exploring alternative spatial interpolation methods (e.g., Kriging) or new downscaling techniques could further enhance the spatial resolution and accuracy of these datasets for small-scale agricultural applications.

To evaluate the reanalysis datasets for irrigation scheduling, we used data from real vineyards in Nemea, allowing for a direct assessment of dataset accuracy under realistic microclimatic conditions. Comparisons were made between in situ meteorological stations at each vineyard and the corresponding grid point estimations from the two reanalysis datasets. This in situ comparison minimized errors that might arise from extrapolating meteorological data to different locations. Other potential sources of uncertainty include instrument errors in the meteorological measurements and inaccuracies inherent in the parameterization of the irrigation needs calculation algorithm. To mitigate these uncertainties, we used data from a relatively new and uniform monitoring network, and all data underwent rigorous quality control for integrity and consistency. Regarding the irrigation needs calculation, we incorporated detailed soil data for the study sites. While some inaccuracies and simplifications in the algorithm are unavoidable, they were applied consistently across all scenarios; therefore, their influence on the relative performance of the datasets is expected to be minimal.

Our study provides important indications of the importance of selecting appropriate datasets and interpolation methods when utilizing global climate datasets for small-scale agricultural applications, particularly in regions with complex topography. AgERA5 demonstrates better performance than MERRA-2 in capturing localized climate variability, making it a more suitable choice for precision agriculture in such areas. However, both datasets have serious limitations, and further research is needed to improve their accuracy and applicability for irrigation scheduling and water management in agriculture.

5. Conclusions

This study assessed the applicability of AgERA5 and MERRA-2 reanalysis datasets for precision irrigation management in the complex, data-scarce viticultural region of Nemea, Greece. While both datasets are widely used in agricultural applications, our findings reveal critical limitations when applied at the local scale in areas with significant spatial variability.

In contrast to the outcomes of previous research in different regions, neither AgERA5 nor MERRA-2 consistently provided the accuracy required for reliable irrigation scheduling in our study area. Although AgERA5 generally outperformed MERRA-2, particularly after

downscaling with elevation-corrected interpolation, both datasets exhibited substantial errors in estimating precipitation and, to a lesser extent, evapotranspiration. These errors, while partially offsetting in the calculation of overall irrigation needs, introduce significant uncertainty into water management decisions.

The discrepancy between our findings and those of studies in other regions underscores that the performance of global climate datasets is highly context-dependent. The uneven performance of global datasets in different regions due to lack of sufficient observation data for reanalysis data calibration was also indicated. This highlights the inherent risk of applying global datasets directly to local agricultural applications without rigorous validation, particularly in regions with similarly complex topography and data scarcity.

These findings have significant implications for sustainable water management, particularly in water-stressed regions. Over- or under-irrigation, resulting from reliance on inaccurate climate data, can lead to reduced grape quality, increased disease susceptibility, and inefficient water use, jeopardizing both economic viability and environmental sustainability.

Future research should prioritize the development of improved downscaling techniques specifically tailored to complex terrain. This could involve integrating high-resolution topographic data, exploring advanced machine learning approaches for bias correction, and incorporating real-time data from local sensor networks, where available. Investigating data fusion methods that combine reanalysis data with satellite-derived products (e.g., for soil moisture) could also offer a promising avenue for improving irrigation scheduling accuracy.

In conclusion, while global reanalysis datasets like AgERA5 and MERRA-2 offer a valuable resource for agricultural applications, their limitations in regions with complex terrain and data scarcity cannot be ignored. Our study serves as a crucial reminder of the need for local validation and the development of context-specific solutions to ensure efficient use of water resources in agriculture.

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Data Availability Statement: AgEra5 and MERRA-2 data are freely available from the corresponding organizations mentioned in the paper. All other data are available by upon reasonable request.

Conflicts of Interest: The authors declare no conflicts of interest.

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